

Jishnu Prasad

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EDUCATION

National Institute of Technology Karnataka (NITK Surathkal)

B.Tech in Electrical and Electronics Engineering CGPA: 7.18 / 10

Surathkal, India

Sep. 2024 – Present

EXPERIENCE

HALE (Healthcare Analytics and Language Engineering) Lab

Research Intern | Edge Computing, ROS2, Raspberry Pi, Ansible

IT Dept., NITK

Jan. 2025 – Present

- Engineered a multi-node **Raspberry Pi 5 edge-computing cluster** for deploying resource-constrained robotics and inference workloads across distributed ARM-based systems.
- Developed robotics workloads using **ROS2**, integrating sensor data, perception pipelines, mapping, and navigation components for edge deployment.
- Built and deployed real-time disaster-intelligence pipelines combining **satellite/drone imagery** with textual data for edge-assisted decision support.
- Automated reproducible provisioning and deployment of edge workloads across Raspberry Pi nodes using **Ansible**, with MetallB, NGINX Ingress, Prometheus, Grafana, Loki, and MinIO.

SPIRE (Signal Processing, Interpretation and REpresentation) Lab

Research Intern | Signal Processing, Multimodal Systems & LLM Evaluation

Indian Institute of Science (IISc)

May 2026 – July 2026

- Developed data-processing pipelines combining **OCR, document processing, web data acquisition, and multimodal retrieval** for English and Kannada banking documentation.
- Worked on retrieval and evaluation pipelines for heterogeneous data sources including PDF, Markdown, Excel, and HTML documents.
- Applied model evaluation and feedback techniques for measuring **accuracy, relevancy, and faithfulness** of generated responses.

EMBEDDED SYSTEMS & ROBOTICS PROJECTS

LLM-Guided Autonomous Rover | ESP32, Raspberry Pi 5, ROS2, YOLO, React Native

- Built an autonomous rover around a **Raspberry Pi 5**, using a webcam-based perception pipeline and ROS2 for robot control and communication.
- Separated high-level computation from low-level actuation by **offloading motor control to an ESP32**, establishing WiFi communication between the embedded controller and the rover interface.
- Implemented a perception-to-command pipeline in which camera frames were processed for navigation decisions and converted into movement commands for the embedded controller.
- Developed a mobile control interface for camera streaming and rover command transmission, with an optional local YOLO-based obstacle-detection path.

Caterpillar Autonomy Challenge — Sand Berm Construction Rover | C++, ROS2, LiDAR, SLAM, Raspberry Pi 5

- Ranked **Top 10 of 1,922 teams** at the IIT Madras challenge while developing an autonomous rover for unstructured outdoor terrain.
- Developed a real-time **LiDAR processing pipeline** using `/scan` data for obstacle detection and avoidance, integrated with SLAM-based occupancy-grid mapping.
- Implemented sensor-fusion and navigation using **LiDAR, odometry, and occupancy grids** on resource-constrained Raspberry Pi 5 + Hailo AI HAT hardware.
- Fine-tuned a YOLOv8n model for crater detection, achieving approximately **92% mAP** for visual perception and obstacle avoidance.

Multi-Camera Live Sky Stitching System | ESP32, Raspberry Pi, MPU6050, OpenCV, PyTorch

- Designed a low-cost multi-camera embedded sensing system using **Raspberry Pi, ESP32, webcams, and an MPU6050 IMU** for orientation-aware image acquisition.
- Developed a synchronized multi-sensor pipeline for camera calibration, orientation estimation, and real-time data processing.
- Implemented homography-based image alignment and real-time panorama generation using OpenCV.

Embedded & Systems Projects | C++, Go, Redis, NATS, Linux

- Memora**: Implemented a Redis-compatible in-memory database in Go with RESP protocol support, TTL expiration, and snapshot persistence, achieving **226K+ ops/sec** at **39 μ s P50** latency.
- ATLAS**: Built a fault-tolerant telemetry and log-aggregation platform with real-time dashboards and a NATS JetStream messaging layer for distributed fleet communication.

TECHNICAL SKILLS

Programming: C++, Python, Go, JavaScript/TypeScript, MATLAB

Embedded & Hardware: ESP32, Raspberry Pi, MPU6050, Hailo AI HAT, Arduino, Sensor Integration, WiFi Communication

Embedded Concepts: Sensor Fusion, Real-Time Data Processing, Edge Computing, Hardware/Software Integration, Control Systems, Signal Processing

Robotics: ROS2, SLAM, LiDAR, Odometry, Occupancy Grids, YOLOv8, OpenCV

Tools & Systems: Linux, Git, Docker, Kubernetes, Ansible, Redis, NATS, Prometheus, Grafana

Engineering: Simulink, MATLAB, Multimodal Signal/Data Processing

LEADERSHIP & ACTIVITIES

Amateur Astronomy Club (AAC) & SMILE Club, NITK

Co-led technical Arduino-based astronomy projects, led technical events, and founded sub-teams to improve project execution and team coordination.

Chief Technical Coordinator

Schneider Electric Go Green 2025 (Team EcoLogic)

Led a team designing a decentralized solar-powered hydrogen and biogas system for sustainable rural energy and waste management.

Team Lead